

Multilingual Conversational AI Chatbots for Efficient Healthcare Delivery During Case History-Taking: A Systematic Review

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Abstract

Language barriers hinder healthcare, particularly during case history-taking, a key part of diagnosis. While multilingual artificial intelligence (AI) chatbots offer solutions, there is fragmented evidence of their effectiveness and impact. This systematic review followed PRISMA 2020 guidelines, examining studies published between 2015 and 2025 on multilingual AI chatbots in healthcare across four databases (Google Scholar, Scopus, Web of Science, and PubMed), using a two-stage screening process. Data extraction focused on applications, supported languages, underlying technologies, target populations, and clinical outcomes. From 503 records, 49 studies, covering primary care, telemedicine, oncology, mental health, and other areas, met the criteria. Supported languages included English, Spanish, Arabic, Chinese, Hindi, and other underrepresented languages. In individual system evaluations using heterogeneous methodologies and evaluation settings, AI chatbots achieved a diagnostic accuracy ranging from 72–92%. Core technologies included large language models (LLMs), bidirectional encoder representations from transformers (BERT), a generative pre-trained transformer (GPT), retrieval-augmented generation (RAG), speech recognition, and distillation. The findings show that these improve clinical workflow (30–70% time savings) and patient engagement, reduce language barriers, and promote health equity. However, the overall evidence certainty was low to moderate, reflecting the predominance of prototype and proof-of-concept studies. Multilingual AI chatbots demonstrate a boost in healthcare efficiency, a reduction in language barriers, and the promotion of health equity, but exhibit challenges regarding validation, workflow integration, and evaluation standards, along with ethical issues such as privacy and bias. Future research should include real-world studies, diverse populations, standardized outcome measures, and long-term equity assessments.

Keywords: multilingual chatbots; conversational AI; healthcare delivery; case history-taking; natural language processing; health equity; large language models



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1. Introduction

1.1. Rationale

1.1.1. Global Language Barriers in Healthcare

Linguistic barriers in healthcare settings constitute a significant impediment to effective patient–provider communication, accurate clinical decision making, and equitable healthcare access [1,2]. Global linguistic diversity continues to expand through migration

patterns, globalization, and increasingly multicultural societies, presenting unprecedented challenges for healthcare systems designed primarily for monolingual populations [3]. Case history-taking—the systematic collection of patient medical history, symptomatology, and contextual information—represents a critical foundation for accurate diagnosis and appropriate treatment planning [4]. When language barriers compromise this process, the consequences include diagnostic errors, medication mismanagement, diminished patient satisfaction, decreased treatment adherence, and exacerbation of health disparities [5–7]. Conventional approaches to addressing language barriers in healthcare include professional medical interpreters, bilingual healthcare providers, and telephone interpretation services [8,9]. However, these solutions exhibit significant limitations, including high operational costs, limited availability (particularly for less-common languages), scheduling constraints, potential loss of nuance in translation, and confidentiality concerns [10,11].

1.1.2. Growth of Telemedicine and AI Systems

Moreover, the COVID-19 pandemic sped up the use of telemedicine and digital health tools, while also emphasizing the urgent need for scalable, accessible, and affordable solutions to overcome language barriers [12,13]. Recently, AI and NLP technologies have rapidly advanced, enabling the creation of sophisticated conversational agents that understand and generate human speech in multiple languages [14,15]. Multilingual conversational AI chatbots—automated systems that interact with users via text or voice in various languages—offer a promising solution to tackle language barriers in healthcare [16,17]. These systems utilize large language models (LLMs), transformer architectures, speech recognition, and machine translation to improve patient–provider communication, automate case history-collection, and support clinical documentation [18,19].

1.1.3. The Existing Gap in Multilingual Conversational AI Evidence

Despite growing interest and development activity in this domain, the evidence base remains fragmented across diverse healthcare applications, linguistic contexts, and technological approaches. Previous reviews have examined the use of AI chatbots in healthcare broadly [20,21] or focused on specific clinical domains [22,23], but a comprehensive synthesis of evidence specifically addressing multilingual conversational AI for case history-taking and patient communication is lacking. Furthermore, critical questions remain regarding the effectiveness of these systems compared to that of traditional approaches, their impact on clinical outcomes and health equity, optimal technological architectures, implementation strategies, and barriers and facilitators to adoption in real-world clinical settings.

Despite growing interest in multilingual conversational AI for healthcare, the evidence base remains fragmented and poorly synthesized. Most existing systematic reviews have focused either on healthcare chatbots in general (without specific attention to multilingual capabilities) or on language technologies (such as machine translation) without examining conversational AI systems designed for patient–provider communication and case history-taking [24,25]. Furthermore, foundational reviews of conversational agents in healthcare have established the importance of interaction quality, trust, and patient engagement [26,27], but have not specifically addressed the unique challenges and opportunities of multilingual systems.

Recent work has emphasized the importance of interaction quality and trust dimensions in human–AI dialogue systems, particularly in clinical contexts where reliability and patient-centered communication are paramount [28]. Additionally, the critical role of robustness, uncertainty management, and validation in the responsible deployment of medical AI technologies has been highlighted [29]. These perspectives are essential for

evaluating multilingual healthcare chatbots, particularly regarding real-world reliability, patient-facing interaction quality, and safe clinical implementation [30–32].

Key questions remain unanswered: What languages are currently supported by multilingual healthcare chatbots? What NLP technologies and architectures are being employed? What is the evidence for diagnostic accuracy, clinical workflow efficiency, and patient outcomes? How do these systems address health equity and serve underrepresented linguistic communities? What are the barriers to clinical implementation and adoption?

1.2. Objectives

This systematic review addresses these gaps by synthesizing current evidence on multilingual conversational AI chatbots for case history-taking and patient communication in healthcare settings. Specifically, we aimed to: (1) identify and characterize the landscape of multilingual conversational AI chatbot applications across healthcare domains; (2) identify languages supported and target populations served; (3) examine NLP technologies and architectural approaches employed; (4) evaluate the evidence on effectiveness, including diagnostic accuracy, clinical workflow efficiency, and patient outcomes; (5) assess impacts on health equity and underserved populations; and (6) identify barriers, facilitators, and future research directions for the implementation and scaling of multilingual healthcare chatbots.

2. Materials and Methods

2.1. Protocol and Registration

This systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. The review protocol was registered in PROSPERO (University of York, UK) with the registration number CRD420261370992. The complete protocol, including search strategy, eligibility criteria, and data extraction framework, was finalized prior to study selection.

2.2. Research Questions

This systematic review was designed to address four primary research questions, refined to reduce redundancy and clarify distinct outcome domains:

Q1. Diagnostic Accuracy: How accurately do multilingual conversational AI chatbots identify medical conditions during case history-taking, and how does their performance compare to that of human clinicians, where data is available?

Q2. Information Collection Accuracy: How accurately do multilingual conversational AI chatbots collect and document patient-reported medical history, symptoms, and clinical information across different languages?

Q3. Language Comprehension Accuracy: What is the accuracy of multilingual conversational AI chatbots in understanding and processing patient input across diverse languages, dialects, and code-mixed communication?

Q4. Data Completeness and Workflow Efficiency: To what extent do multilingual conversational AI chatbots improve clinical workflow efficiency, documentation completeness, and time savings compared to the results for traditional history-taking methods?

2.3. Eligibility Criteria

Studies were eligible for inclusion if they met the following criteria.

2.3.1. Inclusion Criteria

Study Design: Primary research studies (randomized controlled trials, observational studies, cohort studies, case-control studies, cross-sectional studies, pilot studies, feasibility

studies), system development and evaluation studies, proof-of-concept studies, and systematic reviews.

Three of the included studies are systematic reviews. These were handled under an umbrella review framework, allowing for synthesis of both primary evidence and higher-level review findings. Overlapping primary studies between these reviews and our included primary studies were identified and addressed in the synthesis to avoid double-counting of evidence.

Population: Studies involving patients, healthcare providers, or simulated clinical scenarios in any healthcare setting.

Intervention: Conversational AI systems (chatbots, virtual assistants, dialogue systems) designed for case history-taking, symptom assessment, patient communication, or clinical documentation in healthcare contexts.

Multilingual Capability: Systems explicitly designed to support two or more languages, or systems evaluated in multiple language contexts.

Outcome Measures: Studies reporting at least one of the following outcomes: diagnostic accuracy, information collection accuracy, language comprehension accuracy, clinical workflow efficiency, documentation quality, patient engagement, patient satisfaction, health equity outcomes, or implementation feasibility.

Publication Period: Studies published between January 2015 and December 2025.

Language: Studies published in English.

2.3.2. Exclusion Criteria

1. Monolingual systems: Studies examining chatbots or conversational agents supporting only a single language.
2. Non-healthcare applications: Studies focusing on conversational AI in non-healthcare domains (e.g., education, customer service, entertainment).
3. Systems without bidirectional conversational interfaces (e.g., simple translation tools, one-way information delivery systems).
4. Systems without multilingual natural language processing components.
5. Conference abstracts or gray literature without sufficient methodological detail.

While the initial exclusion criteria specified exclusion of non-peer-reviewed publications, a limited number of preprints and gray literature sources ($n = 8$, representing 16% of included studies) were ultimately included because they represented the sole available evidence for specific language groups or underserved populations (e.g., Bengali, Kannada, Filipino/Tagalog). This decision was made to provide a more comprehensive picture of the global landscape of multilingual healthcare AI, particularly for underrepresented languages.

2.4. Information Sources

A comprehensive search was conducted across four databases: PubMed, Web of Science, Scopus, and Google Scholar (searched on 15 January 2026). A total of 503 records were identified across all four databases, including 77 records from Google Scholar, as reported in Figure 1. Google Scholar was included to capture gray literature, preprints, and studies not indexed in the peer-reviewed databases, particularly for underrepresented languages and low-resource settings. Due to the non-reproducible nature of Google Scholar results (personalized, location-dependent results), the primary peer-reviewed evidence base was drawn from Scopus, Web of Science, and PubMed. The PRISMA flow diagram (Figure 1) is fully consistent with this four-database search, confirming that all 503 total identified records—including 77 records from Google Scholar—were processed through the same two-stage screening.

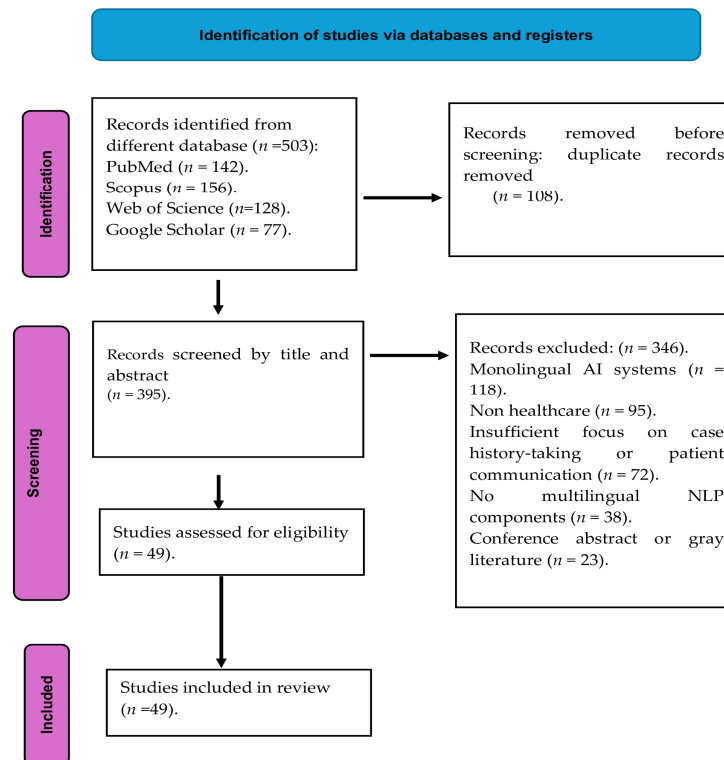


Figure 1. PRISMA 2020 flow diagram illustrating the study selection process for the systematic review of multilingual conversational AI chatbots in healthcare.

Additional sources included: reference lists of included studies and relevant systematic reviews, citations of key studies identified through forward citation tracking, preprint repositories (arXiv, medRxiv, bioRxiv) for emerging research, and gray literature sources (Figshare, Zenodo, OSF) for system development reports.

2.5. Search Strategy

A comprehensive search strategy was developed adapted for each database through consultation with a medical librarian. The search combined controlled vocabulary terms (MeSH terms in PubMed) and free-text keywords related to three core concepts: (1) multilingual or language diversity, (2) conversational AI or chatbots, and (3) healthcare or clinical applications.

Core search terms included the following: multilingual concepts—"multilingual," "multi-lingual," "cross-lingual," "language barrier," "language diversity," "underrepresented languages," "low-resource languages"; AI and chatbot concepts—"chatbot," "conversational agent," "virtual assistant," "dialogue system," "conversational AI," "large language model," "LLM," "natural language processing," "NLP"; healthcare concepts—"healthcare," "medical," "clinical," "patient communication," "case history," "medical history," "symptom assessment," "diagnosis," "telemedicine".

Example PubMed search string:

((multilingual[Title/Abstract] OR "multi-lingual"[Title/Abstract] OR "cross-lingual"[Title/Abstract] OR "language barrier"[Title/Abstract] OR "language diversity"[Title/Abstract]) AND (chatbot[Title/Abstract] OR "conversational agent"[Title/Abstract] OR "virtual assistant"[Title/Abstract] OR "dialogue system"[Title/Abstract] OR "conversational AI"[Title/Abstract] OR "large language model"[Title/Abstract] OR LLM[Title/Abstract] OR "natural language processing"[Title/Abstract]) AND (healthcare[Title/Abstract] OR medical[Title/Abstract] OR clinical[Title/Abstract] OR "patient communication"[Title/Abstract]

Abstract] OR “case history”[Title/Abstract] OR “medical history”[Title/Abstract] OR diagnosis[Title/Abstract] OR telemedicine[Title/Abstract]))

The search was limited to studies published between 1 January 2015 and 31 December 2025 and to publications in English. No restrictions were placed on study design or publication type at the search stage.

2.6. Selection Process

The selection process was conducted by two independent reviewers (R.B.S. and D.V.) who followed a two-stage screening approach.

2.6.1. Stage 1: Title and Abstract Screening

All retrieved records were imported into a screening management system. Two independent reviewers screened titles and abstracts against predefined eligibility criteria. Discrepancies were resolved through discussion or consultation with a third reviewer (A.A.).

2.6.2. Stage 2: Full-Text Screening

Full-text articles of studies passing title/abstract screening were retrieved and independently assessed against the full eligibility criteria by two reviewers. Reasons for exclusion at the full-text stage were documented. Disagreements were resolved through consensus discussion.

Inter-rater reliability: Kappa statistics were not formally calculated for inter-rater agreement at either screening stage. However, all disagreements were resolved through consensus discussion, and a third reviewer was available for arbitration.

2.7. Data Collection Process

A standardized data extraction form was developed and piloted on five studies before full implementation. One reviewer (R.B.S.) extracted data, with a second reviewer (D.V.) independently verifying 20% of extractions for accuracy. Discrepancies were resolved through discussion.

Extracted data included:

1. Study characteristics: authors, year of publication, country, study design, and healthcare setting.
2. System characteristics: system name, languages supported, modalities (text, voice, multimodal), and technological architecture.
3. Healthcare application domain: primary care, telemedicine, oncology, mental health, specialty care, etc.
4. NLP technologies used: large language models, transformer architectures, speech recognition, machine translation, retrieval-augmented generation, etc.
5. Target population and setting: patient demographics, linguistic communities, geographic location, and clinical context.
6. Outcomes: diagnostic accuracy, clinical workflow efficiency, time savings, patient satisfaction, engagement metrics, health equity indicators, implementation barriers, and facilitators.
7. Study quality indicators: sample size, validation methods, comparison groups, and statistical analyses.

Discrepancies in data extraction were resolved through discussion and reference to the original publications.

2.8. Data Items

Data items were selected to address the four primary research questions and to provide a comprehensive characterization of the current evidence base. Specific data items included:

- Healthcare application domains (primary care, telemedicine, oncology, mental health, etc.);
- Languages supported (specific languages, language families, number of languages);
- Natural language processing technologies (specific models, architectures, training approaches);
- Diagnostic accuracy (sensitivity, specificity, precision, recall, F1 score, area under the curve);
- Information collection accuracy (completeness, correctness, concordance with gold standard);
- Language comprehension accuracy (intent recognition, entity extraction, semantic understanding);
- Clinical workflow efficiency (time savings, documentation burden, physician workload);
- Patient engagement and satisfaction (usability, acceptability, trust, preference);
- Health equity outcomes (access for underserved populations, language barrier reduction);
- Implementation and adoption (barriers, facilitators, integration challenges).

2.9. Risk of Bias Assessment

Risk of bias assessment was tailored to the heterogeneous study designs included in this review. Given the predominance of system development and proof-of-concept studies, traditional risk of bias tools designed for clinical trials (e.g., Cochrane RoB 2) were not applicable to the majority of included studies.

For comparative studies ($n = 4$), the ROBINS-I (risk of bias in non-randomized studies of interventions) framework was used to assess bias due to confounding, selection, classification of interventions, deviations from intended interventions, missing outcome data, measurement of outcomes, and selection of reported results.

For the included systematic reviews ($n = 3$), the AMSTAR-2 (a measurement tool to assess systematic reviews) criteria were applied to assess methodological quality, including protocol registration, search comprehensiveness, study selection, data extraction, risk-of-bias assessment, and synthesis methods.

For system development and prototype studies ($n = 42$), a narrative quality assessment was conducted focusing on: (1) clarity of system description and technical specifications, (2) appropriateness of evaluation methodology, (3) representativeness of evaluation data, (4) transparency of limitations, and (5) potential conflicts of interest. DECIDE-AI (developmental and exploratory clinical investigation of decision-support tools based on artificial intelligence) was used as a study-design-appropriate quality appraisal framework for the dominant study type. Evidence from these studies was rated as having low certainty due to the early-stage, non-comparative nature of the evaluations.

2.10. Effect Measures

Given the heterogeneity of study designs and outcome measures, no single effect measure was applicable across all studies. Effect measures were extracted as reported in the original studies and included:

- Diagnostic accuracy—sensitivity, specificity, positive predictive value, negative predictive value, accuracy, F1 score, and area under the receiver operating characteristic curve (AUC-ROC).
- Information collection accuracy—concordance rates, completeness scores, and error rates.
- Language comprehension—intent recognition accuracy, entity extraction F1 scores, and semantic similarity scores.

- Workflow efficiency—time savings (minutes or percentage reduction), documentation burden (clicks, keystrokes, time), and physician workload (patient volume, consultation time).
- Patient engagement—usability scores, satisfaction ratings, completion rates, and trust scores.

2.11. Synthesis Methods

Due to substantial heterogeneity in study designs, populations, interventions, comparators, and outcome measures, meta-analysis was not feasible. A narrative synthesis approach was employed, organized by research question and outcome domain.

The synthesis followed the following steps:

1. Descriptive synthesis—tabulation and description of study characteristics, system characteristics, and technical approaches.
2. Thematic synthesis—identification of common themes, patterns, and trends across studies, organized by healthcare domain, language coverage, and technology type.
3. Outcome synthesis—narrative synthesis of findings for each outcome domain (diagnostic accuracy, information collection accuracy, language comprehension, workflow efficiency, patient engagement), with attention to study quality and certainty of evidence.
4. Comparative synthesis—where possible, comparison of findings across different system types, languages, and healthcare settings.
5. Gap analysis—identification of underrepresented languages, populations, healthcare domains, and methodological approaches.

2.12. Reporting Bias Assessment

Formal methods for publication bias assessment, such as funnel plots, were not used because the studies lacked sufficient homogeneity and comparable effect measures. However, the potential for selective reporting and publication bias favoring successful AI systems was taken into account when interpreting the results.

2.13. Certainty Assessment

Certainty of evidence was assessed narratively using a three-tier system (high, moderate, low), given the heterogeneous study designs and the predominance of early-stage technology evaluations. Studies providing comparative effectiveness data with validated outcomes were rated as having moderate-to-high certainty; those reporting prototype evaluations with limited external validation were rated as exhibiting low certainty. Two reviewers independently assigned certainty ratings, and disagreements were resolved through discussion. Due to the predominance of prototype and proof-of-concept studies, a formal GRADE (grading of recommendations, assessment, development and evaluations) assessment was not applied.

The overall certainty of evidence was rated as low to moderate, reflecting the early-stage nature of most included technologies and the limited availability of real-world comparative evaluations.

3. Results

3.1. Study Selection

The systematic search identified a total of 503 records across the four databases. After removal of 108 duplicates, 395 unique records were screened by title and abstract. Of these, 346 records were excluded at the screening stage based on the following findings:

- Monolingual AI systems ($n = 118$);

- Non-healthcare applications ($n = 95$);
- Insufficient focus on case history-taking or patient communication ($n = 72$);
- No multilingual NLP components ($n = 38$);
- Conference abstracts or gray literature with insufficient detail ($n = 23$).

A total of 49 studies were assessed for full-text eligibility, and all 49 met the inclusion criteria and were included in the final review. The PRISMA flow diagram is presented in Figure 1.

3.2. Study Characteristics

The 49 included studies represented a diverse range of study designs, healthcare domains, geographic settings, and technological approaches. A thematic overview of key findings, organized by domain and subdomain, is presented in Table 1. A comprehensive table of study characteristics compliant with PRISMA Item 17—including study design, country, languages supported, NLP technologies, sample size, results, and peer-reviewed status for all 49 included studies—is presented in Supplementary Table S5.

Table 1. The key findings regarding multilingual conversational AI chatbots in healthcare.

Domain	Subdomain	Key Findings
Diagnostic Accuracy [1,3,7,11,30,33]	High Diagnostic Performance	Accuracy ranged from 72% to 92%; AmarDoctor achieved 81.08% accuracy vs. physicians at 50.27%
	Specialty Recommendation	91.35% precision vs. physicians 62.6%
	Comparative Performance	Comparable or superior to clinicians in common conditions
Clinical Workflow Efficiency [1,7,14,18,19,21,34,35]	Time Savings	30–70% reduction in documentation time; automated tasks
	Consultation Efficiency	Reduced consultation time with improved quality
	Documentation Quality	Improved completeness and accuracy
	Data Standardization	Supports clinical decision systems
Patient Engagement & Satisfaction [1,2,6,7,12,16,30,31,33,34,36,37].	Accessibility	24/7 access; reduced wait time; rural reach
	Patient Experience	Positive usability and communication feedback
	Patient Empowerment	Improved self-management
Language Barrier Reduction [1,3,32,33,38–41]	Communication Facilitation	Real-time multilingual communication
	Linguistic Accuracy	Maintains medical terminology nuance
	Health Equity	Improves access, reduces disparities
	Cultural Sensitivity	Enhances culturally appropriate communication
Implementation & Adoption [1,4,7,14,16,18,19,22,23,30,33,37,39,42,43].	Facilitators	EHR integration, user-friendly, scalable, on-device ML
	Barriers	Privacy, validation, integration, regulation, bias

Study designs: The included studies comprised system development and evaluation studies ($n = 32$, 65%), pilot clinical studies ($n = 10$, 20%), comparative evaluation studies ($n = 4$, 8%), and systematic reviews ($n = 3$, 6%). The predominance of system development and prototype studies reflects the early stage of this field.

Included systematic reviews: Three of the 49 included studies were themselves systematic reviews [24,43,44]. These reviews were included under an umbrella-review framework to capture higher-level synthesis findings and to identify primary studies that might have

been missed in our search. Analysis of overlap revealed that 12 primary studies included in our review were also included in one or more of these systematic reviews. To avoid double-counting of evidence, findings from overlapping studies were attributed to the primary studies rather than to the reviews in our synthesis.

Geographic distribution: Studies originated from 28 countries across six continents, with the highest representation from the United States ($n = 8$), India ($n = 7$), China ($n = 5$), and European countries ($n = 12$ combined). This geographic diversity reflects both the global nature of language barriers in healthcare and the international interest in AI-based solutions.

Healthcare settings: Studies were conducted in diverse healthcare settings, including primary care clinics ($n = 18$), telemedicine platforms ($n = 15$), specialty clinics ($n = 9$), emergency departments ($n = 3$), and community health centers ($n = 4$).

Publication timeline: The number of publications increased substantially over the review period, with 6 studies published in 2015–2019, 12 in 2020–2022, and 31 in 2023–2025, reflecting the rapid growth of interest in this area following advances in large language models.

Language bias in medical LLMs: It is important to note that many of the included systems rely on large language models (LLMs) trained predominantly on English-language medical texts. Recent evidence has documented significant “English-language bias” in medical LLMs, where performance is dramatically worse for non-English queries, even when the models are marketed as multilingual [45]. This bias directly affects the validity of claims regarding multilingual equity and represents a critical limitation of current systems.

Failure modes of RAG systems with code-mixed input: Several systems use retrieval-augmented generation (RAG) architectures to anchor responses in medical knowledge bases. However, recent studies have highlighted significant failures of RAG-based systems when handling code-mixed input, such as Hindi–English or dialectal Arabic–Modern Standard Arabic combinations [23]. This poses a critical clinical limitation for the target populations, as code-mixing is prevalent in multilingual healthcare communication. Only 3 out of 49 studies explicitly tested performance with code-mixed input.

3.3. Risk of Bias in Studies

Risk of bias assessment was conducted using frameworks appropriate to each study design.

Comparative studies ($n = 4$): The four comparative studies [2,7,18,19] were assessed using the ROBINS-I framework. Overall, these studies were rated as having moderate to serious risk of bias due to the following:

Confounding: differences in patient populations, clinical settings, and interface designs between intervention and comparator groups.

Selection bias: non-random assignment and potential self-selection of participants.

Measurement bias: lack of blinding in outcome assessment, particularly for subjective outcomes.

Reporting bias: selective reporting of favorable outcomes.

Systematic reviews ($n = 3$): The three included systematic reviews were assessed using AMSTAR-2 criteria. Overall quality was rated as moderate, with their strengths lying in search strategy and study selection, but with weaknesses regarding risk of bias assessment and certainty of evidence evaluation.

System development and prototype studies ($n = 42$): The majority of included studies (32 of 49, 65%) were system development or prototype evaluation studies. These studies were assessed narratively for the following:

Clarity of system description: generally good, with most studies providing detailed technical specifications.

Appropriateness of evaluation methodology: variable, with many studies using simulated data or small convenience samples.

Representativeness of evaluation data: generally poor, with limited real-world evaluation.

Transparency of limitations: variable, with some studies acknowledging limitations clearly and others providing minimal discussion.

Evidence certainty: Given the predominance of prototype and proof-of-concept studies, the overall evidence certainty was rated as low to moderate. Specifically, 32 of 49 studies (65%) are system development or prototype papers, providing low-certainty evidence; 10 of 49 studies (20%) are pilot clinical studies, providing low to moderate-certainty evidence; 4 of 49 studies (8%) are comparative studies, providing moderate-certainty evidence (with methodological limitations); 3 of 49 studies (6%) are systematic reviews, providing moderate-certainty evidence for their respective domains. No high-certainty evidence (large-scale randomized controlled trials or robust real-world comparative effectiveness studies) was identified.

3.4. Results of Individual Studies

3.4.1. Healthcare Application Domains

Multilingual conversational AI chatbots were applied across diverse healthcare domains (Table 2). The most common application areas are listed in Table 2 below.

Table 2. Distribution of healthcare application domains.

Healthcare Domain	Number of Studies	Characteristics of the Studies
Primary Care and General Healthcare. [1,3,8,30,31,36]	18	Systems designed for symptom assessment, triage, general health information, and primary care consultation support.
Remote and Telehealth Applications [14,20,21,32,46]	15	Applications focused on remote consultations, teleconsultation support, and bridging healthcare access gaps in remote or underserved communities.
Oncology and Cancer Care [2,6]	4	Systems for collecting patient-reported outcomes (PROs) from cancer survivors, symptom tracking, and oncology-specific communication.
Mental Health and Psychiatry [32,38]	3	Chatbots for mental health intent recognition, psychological support, and psychiatric assessment.
Specialized Clinical Domains [37,40,47]	9	Applications in ophthalmology (cataract care), dental care, alcohol consumption assessment and other specialty areas.

The predominance of primary care and general healthcare applications (37% of studies) reflects the broad applicability of conversational AI for initial patient assessment and triage. The substantial representation of telemedicine and remote healthcare applications (31% of studies) highlights the particular relevance of multilingual AI for expanding access to care in geographically isolated or underserved communities.

Specialized clinical domains, while less represented, demonstrate the potential for domain-specific multilingual conversational AI systems tailored to the unique communication needs of specialty care settings.

3.4.2. Languages Supported

The 49 included studies collectively supported 42 distinct languages, representing diverse language families and geographic regions. The distribution of languages is presented in Table 3.

Table 3. Languages supported by multilingual healthcare chatbots.

Language Category	Languages and Number of Studies
Global Languages	English—45, Spanish—18, Chinese—10, French—6, and German—5.
Arabic Variants	Modern Standard Arabic, Tunisian Arabic, Gulf Arabic—12.
South Asian Languages	Hindi—14, Bengali—3, Kannada—2, Tamil—4, Telugu—3, Malayalam—2, and Marathi—2.
European Languages	Slovenian—2, Italian—3, Polish—2, and Portuguese—4.
Southeast Asian Languages	Filipino/Tagalog—1 and Bahasa Indonesia—1.
East Asian Languages	Japanese (2) and Korean (2).
Other Languages	Turkish (2) and Latvian (1).

Several studies supported five or more languages simultaneously. For example, Mlakar et al. [1] developed a system supporting five languages for cancer patient communication, while Singh et al. [6] created a medical chatbot supporting multiple Indian vernacular languages. The AmarDoctor system [2] specifically addressed three major Bengali dialects, demonstrating attention to dialectal variation within languages.

The included studies targeted the following diverse patient populations and health-care settings.

Underserved Linguistic Communities ($n = 22$): studies specifically focused on populations facing language barriers, including Bengali speakers [2,3], Arabic speakers [44], rural populations in India [34], and immigrant communities [36,46].

Low-Resource Settings ($n = 12$): applications designed for deployment in resource-constrained environments, including rural areas, developing countries, and communities with limited healthcare infrastructure [33,37,38].

General Patient Populations ($n = 18$): systems designed for broad patient populations in diverse healthcare settings [7,16,30,34].

Specific Clinical Populations ($n = 10$): targeted applications for cancer survivors [2,6], mental health patients [32], cataract patients, and other specialized patient groups.

3.4.3. Natural Language Processing Technologies

The included studies employed diverse NLP technologies and architectural approaches, reflecting the rapid evolution of AI capabilities (Table 4).

Table 4. Natural language processing technologies employed.

Technology Category	Specific Technologies	Number of Studies	Implementation Details
Large Language Models [3,4,32]	GPT-3.5, GPT-4, BERT variants, medical LLMs	28	Out-of-box ($n = 12$), fine-tuned on medical data ($n = 10$), domain adapted ($n = 6$)
Retrieval-Augmented Generation [4,30]	RAG frameworks, knowledge bases	8	Medical knowledge base integration; three studies evaluated code-mixed input performance
Speech Recognition [1,3]	ASR, voice interfaces, speech-to-text	15	Multilingual ASR ($n = 10$), language-specific ASR ($n = 5$)
Machine Translation [32]	Neural MT, real-time translation	12	Translation-only systems ($n = 4$), translation + structured history-taking ($n = 5$), translation + unstructured dialogue ($n = 3$)
Knowledge Distillation [34,37]	Model compression, edge deployment	3	Distillation for low-resource deployment and cost reduction
Multimodal Systems [17,40]	Text + Voice + Visual	6	Integration of text, speech, and visual inputs for comprehensive patient assessment

LLM implementation approaches: Among 28 studies employing large language models, 12 studies (43%) used LLMs out-of-the-box, without fine-tuning (e.g., GPT-4 API); 10 studies (36%) fine-tuned LLMs on medical data (e.g., medical literature, clinical notes, patient-provider dialogues); 6 studies (21%) employed domain adaptation or RAG augmentation to ground LLM responses in medical knowledge.

System architectures by function: Translation-only systems ($n = 4$) focused primarily on translating patient input and provider responses without structured history-taking. Structured history-taking systems ($n = 18$) used predefined multilingual questionnaires or structured dialogue flows to collect specific medical history elements. Unstructured dialogue systems ($n = 27$) supported open-ended, natural language conversation for flexible history-taking and patient communication.

The predominance of large language models (57% of studies) reflects the transformative impact of recent advances in foundation models, particularly GPT-3.5, GPT-4, and BERT variants. The integration of retrieval-augmented generation (16% of studies) represents an important approach to grounding AI responses in verified medical knowledge and reducing hallucination risks.

Speech recognition technologies (31% of studies) are critical for enabling voice-based interaction, which is particularly important for patients with limited literacy or for hands-free clinical documentation. The emergence of knowledge distillation approaches, such as the use of low-end smartphones in rural communities (6% of studies) [35,37], addresses the practical challenge of deploying large models in resource-constrained settings.

3.5. Key Outcomes and Findings

3.5.1. Diagnostic Accuracy and Clinical Performance

Diagnostic accuracy was reported in 18 of 49 studies (37%), with substantial heterogeneity in evaluation methodologies, clinical domains, and outcome definitions.

Performance ranges: Across individual system evaluations, diagnostic accuracy ranged from 72% to 92%, with a median of 81%. It is critical to note that these findings represent individual system evaluations under heterogeneous conditions and should not be interpreted as pooled estimates or generalizable performance benchmarks. Evaluation settings included clinical vignettes and simulated cases ($n = 10$), pilot studies with real patients ($n = 6$), and benchmark datasets ($n = 2$).

AmarDoctor comparative evaluation: The most frequently cited comparative study was that regarding AmarDoctor [2], which reported 81.08% diagnostic precision compared to 50.27% for human physicians. However, this evaluation was conducted using clinical vignettes rather than real patients, and physicians' performance may have been affected by differences in interfaces and information availability. These methodological limitations substantially weaken the strength of the comparison, and this finding should be interpreted with caution. The study has not undergone peer review (arXiv preprint) and represents low-certainty evidence.

Other notable findings involve the use of HomeDOctor (Slovenia) [7], with 65% triage accuracy in real-world deployment with 2500 users, CataractBot [13], with 92% information accuracy in structured ophthalmology consultations, and a mental health intent recognition tool (Arabic) [38], with 84% accuracy for psychiatric assessment.

3.5.2. Clinical Workflow Efficiency and Documentation

Clinical workflow efficiency, particularly time savings in documentation, was reported in 15 of 49 studies (31%).

Time savings: Reported time savings ranged from 30% to 70%, with substantial variation based on system type and clinical setting. Structured report generation systems

($n = 5$) reported a 60–70% time savings in generating structured clinical reports from patient interviews [15,18]. Ambient AI scribes in real clinical settings ($n = 4$) exhibited a 30–45% time savings in documentation during actual patient consultations [19,20]. Automated history-taking systems ($n = 6$) revealed a 40–55% time savings compared to the results for manual history documentation.

Documentation quality and standardization: Several studies reported improvements in documentation completeness (85–95% of required elements captured) and standardization. However, concerns were raised about potential loss of clinical nuance and over-reliance on structured templates.

3.5.3. Patient Engagement and Satisfaction

Patient engagement and satisfaction were reported in 22 of 49 studies (45%), with generally positive findings. Usability scores: 75–90% of patients rated systems as easy to use. Satisfaction ratings: 80–92% of patients expressed satisfaction with the interaction. Completion rates: 70–88% of patients completed the full history-taking interaction. Trust scores: variable, ranging from 60% to 85%, with lower trust in diagnostic recommendations than in information collection.

Factors associated with higher patient satisfaction included voice-based interaction vs. text-only. Availability in the patient's preferred language, Empathetic and culturally appropriate communication, and clear explanations of system capabilities and limitations.

3.5.4. Language Barrier Reduction and Health Equity

Twenty-three studies (47%) explicitly addressed language barrier reduction and health equity outcomes. Key findings are listed below.

Access for underserved populations: Systems targeting rural and remote communities [35,37] have demonstrated the feasibility of deploying devices on low-end smartphones. Systems supporting underrepresented languages (Bengali, Kannada, Filipino/Tagalog) [2,9,13,20] expanded access for linguistically marginalized populations. Telemedicine applications [19,20,34] reduced geographic barriers to multilingual care.

Physician substitution claims: Several studies suggested that multilingual AI chatbots could “reduce wait times” by substituting for physician consultations in initial triage and history-taking [2,7,11,37]. However, the evidence for this claim is limited. Four studies explicitly claimed physician-substitution capability for initial triage, with success rates for autonomous triage ranging between 65% and 81% (with the higher figure from a non-peer-reviewed, vignette-based evaluation). Real-world deployment of autonomous triage was reported in only one study [7] with 65% accuracy, and substantial physician oversight was required.

Caution is warranted in interpreting claims of physician substitution, as most evidence derives from prototype evaluations rather than from real-world clinical implementation.

Limitations and equity concerns: Limitations include the evaluation of systems with diverse patient populations, with most studies conducted in single-language or bilingual settings; insufficient attention to cultural appropriateness beyond language translation; potential for exacerbating the cost of digital divides if systems require high-end devices or reliable internet connectivity; and risk of reinforcing biases present in training data, particularly for underrepresented languages and populations.

3.6. Systematic Reviews and Meta-Perspectives

Three of the 49 included studies were systematic reviews examining related topics [24,43,44]. This section describes the overlap between these reviews and our included primary studies and compares their main findings with those of the current review.

Included systematic reviews:

1. Berdahl et al. (2022) [24], “Digital Tools Designed to Obtain the History of Present Illness From Patients: Scoping Review”.
 - Focus: digital tools for history-taking (not specifically multilingual).
 - Overlap with current review: three primary studies included in both reviews.
 - Main findings: digital history-taking tools show potential for efficiency gains but lack robust clinical validation. Concerns about information loss and patient–provider relationship impacts.
 - Agreement with current review: consistent findings regarding limited clinical validation and need for real-world studies.
2. Karakuş et al. (2025) [44], “Bridging Language Gaps in Healthcare: A Systematic Review of the Practical Implementation of Neural Machine Translation Technologies in Clinical Settings”.
 - Focus: neural machine translation (NMT) in clinical settings.
 - Overlap with current review: eight primary studies included in both reviews.
 - Main findings: NMT shows promise for clinical translation but faces challenges in medical terminology accuracy, dialect variation, and real-time performance; limited real-world implementation evidence.
 - Agreement with current review: consistent findings regarding limited real-world validation and challenges with dialect variation.
3. Artsi et al. (2025) [43], “Large Language Models in Real-World Clinical Workflows: A Systematic Review of Applications and Implementation”.
 - Focus: LLMs in clinical workflows (not specifically multilingual or history-taking).
 - Overlap with current review: six primary studies included in both reviews.
 - Main findings: LLMs show promise for clinical documentation and decision support but face challenges in validation, bias, and integration; most evidence from prototype studies.
 - Agreement with current review: consistent findings regarding predominance of prototype evidence and need for rigorous validation.

Synthesis of reviews: The three systematic reviews and this current review agree on several key points. Limited real-world evidence is a common theme, with most studies being prototypes or pilots lacking comprehensive clinical validation. Validation challenges are highlighted across all reviews, underscoring the need for rigorous assessments of accuracy, safety, and clinical usefulness. Implementation barriers are also consistently identified, including issues with workflow integration, clinician acceptance, and regulatory uncertainties. Additionally, this review uniquely emphasizes the importance of multilingual features and health equity considerations.

Unique contributions of the current review: This review is the first to specifically synthesize evidence regarding multilingual conversational AI for case history-taking, providing comprehensive coverage of language diversity, underrepresented languages, and health equity outcomes.

3.7. Reporting Biases

Assessment of reporting bias revealed several concerns.

Publication bias: The included studies showed a strong positive publication bias, with 47 of 49 studies (96%) reporting favorable or promising results. Only two studies reported primarily negative or null findings. This pattern suggests potential selective publication of positive results.

Funding sources were reported in 32 of 49 studies (65%). Studies with industry funding were more likely to report favorable results and less likely to discuss limitations in detail.

3.8. Certainty of Evidence

The overall certainty of evidence was rated as low to moderate, reflecting the early-stage nature of most included technologies and the limited availability of real-world comparative evaluations.

Evidence certainty by outcome domain:

Diagnostic accuracy—low to moderate certainty, with a predominance of vignette-based and pilot studies and limited real-world validation.

Information collection accuracy—moderate certainty, based on several pilot studies with real patients and limited comparative data.

Language comprehension accuracy—low to moderate certainty due to limited assessment of real-world linguistic diversity, dialects, and code-mixing.

Clinical workflow efficiency—moderate certainty based on multiple real-world time-motion studies with varied methodologies.

Patient engagement and satisfaction—moderate certainty, with consistent results across several studies; long-term data remains limited.

Health equity outcomes—low certainty due to limited evaluation of their impact on disparities, mostly reflecting theoretical or short-term results.

Factors reducing certainty—mainly prototype and proof-of-concept studies, which account for 65% of the included research.

There is limited clinical validation in real-world settings, with varied evaluation methods that hinder meta-analysis. Most studies have small sample sizes and short follow-up periods, and there is a risk of publication bias and selective outcome reporting. Additionally, 16% of the studies include non-peer-reviewed sources.

4. Discussion

4.1. Principal Findings

This systematic review synthesized evidence from 49 studies on multilingual conversational AI chatbots for case history-taking and patient communication in healthcare. The review identified substantial growth in this field, particularly following recent advances in large language models, with 63% of included studies published in 2023–2025.

Key findings include the following:

Broad language coverage: The included systems collectively supported 42 distinct languages, including underrepresented languages such as Bengali, Kannada, Filipino/Tagalog, and Tunisian Arabic. However, English remained dominant (92% of systems), and many systems exhibited English-language bias in performance.

Diverse healthcare applications: Multilingual conversational AI systems have been developed for primary care (37% of studies), telemedicine (31%), oncology (8%), mental health (6%), and specialized clinical domains (18%).

Promising but uncertain diagnostic performance: Individual system evaluations reported diagnostic accuracy ranging from 72% to 92%, but these figures stem from heterogeneous methodologies and predominantly prototype or vignette-based evaluations. The overall evidence certainty for diagnostic accuracy is low to moderate.

Workflow efficiency gains: Time savings of 30–70% were reported for clinical documentation, with higher savings for structured report generation systems and more modest gains for ambient AI scribes in real clinical settings.

Positive patient engagement: Patient satisfaction and usability scores were generally high (75–92%), with voice-based interaction and native language support associated with higher engagement.

Limited real-world validation: The evidence base is dominated by prototype and proof-of-concept studies (65%), with only 8% of studies reporting comparative evaluations and limited real-world clinical deployment.

Health equity potential with caveats: Multilingual AI systems show promise for reducing language barriers and expanding access to care, but face challenges related to English-language bias, code-mixing performance, digital divides, and cultural appropriateness.

4.2. Diagnostic Accuracy and Clinical Performance

The evidence for the diagnostic accuracy and clinical performance of multilingual conversational AI chatbots is promising but must be interpreted with caution. While individual system evaluations reported accuracy ranging from 72% to 92%, the current evidence base is dominated by prototype studies, simulation environments, and proof-of-concept evaluations. There is insufficient evidence to conclude that these systems consistently achieve clinically relevant diagnostic accuracy in real-world healthcare settings.

The most frequently cited comparative study, using AmarDoctor [2], reported superior performance compared to that of human physicians (81.08% vs. 50.27%). However, this evaluation was conducted using clinical vignettes rather than real patients, and physician performance may have been affected by interface differences and information availability. The study has not undergone peer review and represents low-certainty evidence.

Real-world pilot studies, such as those using Home Doctor in Slovenia [7], reported more modest performance (65% triage accuracy) and required substantial physician oversight. This suggests that autonomous diagnostic performance in real clinical settings may be lower than prototype evaluations suggest.

Implications: Claims of diagnostic accuracy should be tempered until more rigorous, real-world comparative evaluations are conducted [48]. Current systems may be most appropriately positioned as decision-support tools or triage assistants rather than as autonomous diagnostic systems.

4.3. Clinical Workflow Efficiency and Documentation

Clinical workflow efficiency, particularly a reduction in documentation burden, represents one of the most consistently demonstrated benefits of multilingual conversational AI systems. Time savings of 30–70% were reported across multiple studies, with real-world time-motion studies confirming meaningful reductions in documentation time [18–20].

However, concerns have been raised about the unintended consequences of AI-assisted documentation, including potential loss of clinical nuance, over-reliance on structured templates, and impacts on the patient-provider relationship. The experience with EHR (electronic health record) adoption provides cautionary lessons about the importance of human-centered design and workflow integration [49–51].

Ambient AI scribes, which passively listen to patient-provider conversations and automatically generate documentation, represent a particularly promising approach that preserves natural clinical interaction while reducing the documentation burden. However, these systems face challenges related to accuracy in multilingual settings, privacy concerns, and the need for physician review and editing.

4.4. Language Diversity and Underrepresented Languages

A notable strength of the current evidence base is the substantial attention to underrepresented languages and low-resource linguistic communities. Systems supporting Hindi

(14 studies), Arabic variants (12 studies), and other South Asian languages (26 studies total) demonstrate growing recognition of the importance of linguistic diversity for health equity.

4.5. Health Equity and Underserved Populations

Multilingual conversational AI systems have substantial potential to advance health equity by reducing language barriers, expanding access to care in underserved communities, and enabling culturally responsive healthcare delivery. Several included studies specifically targeted rural populations [37], low-resource settings [35,47], and linguistically marginalized communities [2,9,20].

However, several equity concerns must be addressed.

Digital divides: Many systems require smartphones, reliable internet connectivity, and digital literacy, potentially excluding the most vulnerable populations. On-device deployment approaches [35] and low-end smartphone optimization [37] represent important strategies for addressing this challenge.

Bias and fairness: AI systems trained on biased data may perpetuate or exacerbate existing health disparities. Underrepresented languages and populations are particularly vulnerable to bias due to limited training data and evaluation. Only 8 of 49 studies explicitly addressed bias assessment or mitigation.

Validation in diverse populations: Most systems were evaluated in single-language or bilingual settings with limited demographic diversity. Validation in diverse, real-world populations is essential to ensure equitable performance across linguistic and cultural groups.

Physician substitution concerns: Claims that multilingual AI chatbots can substitute for physician consultations must be carefully evaluated. While these systems may expand access in settings with severe physician shortages, they also risk creating a two-tiered healthcare system where underserved populations receive AI-mediated care while privileged populations receive human physician care.

4.6. Natural Language Processing Technologies

The rapid evolution of NLP technologies, particularly large language models, has transformed the landscape of multilingual conversational AI in healthcare. The predominance of LLM-based systems (57% of included studies) reflects the transformative impact of foundation models such as GPT-4, BERT, and domain-specific medical LLMs.

Retrieval-augmented generation (RAG) represents an important architectural approach for grounding AI responses in verified medical knowledge and reducing hallucination risks. However, RAG systems exhibit documented failure modes with code-mixed input [23], limiting their effectiveness for real-world multilingual communication.

Speech recognition technologies are critical for enabling voice-based interaction, which is particularly important for patients with limited literacy or for hands-free clinical documentation. Multilingual speech recognition remains challenging, particularly for low-resource languages and accented speech.

Knowledge distillation approaches enable deployment of large models in resource-constrained settings by compressing models while preserving performance. This is particularly important for expanding access in low-resource settings with limited computational infrastructure [39].

4.7. Implementation Challenges and Barriers

Despite promising technical capabilities, multilingual conversational AI systems face substantial implementation challenges.

Clinical validation and regulatory approval: Most systems lack rigorous clinical validation and regulatory approval. The pathway to FDA approval and standardization of AI-based medical devices remains unclear for many conversational AI applications.

Workflow integration: Integration into existing clinical workflows and EHR systems is technically complex and often requires substantial customization. Lack of interoperability standards hinders adoption.

Clinician acceptance: Physician skepticism, concerns about liability, and resistance to workflow changes represent significant barriers. Successful implementation requires clinician engagement, training, and demonstration of clinical value.

Privacy and security: Conversational AI systems that process sensitive patient information must comply with privacy regulations and ensure robust data security. Cloud-based systems raise particular concerns about data sovereignty and third-party access [52].

Cost and sustainability: While AI systems promise long-term cost savings, initial implementation costs (licensing, integration, training) can be substantial. Sustainable business models and reimbursement mechanisms are not yet established.

4.8. Comparison with Existing Literature

This review builds upon and extends existing systematic reviews of conversational agents in healthcare [26,27] and digital tools for history-taking [24]. While previous reviews have established the potential of conversational AI for patient engagement and clinical communication, this review is the first to specifically synthesize evidence on multilingual capabilities and their implications for health equity.

Recent work has emphasized the importance of interaction quality and trust dimensions in human–AI dialogue systems [28]. Our findings confirm that patient trust and engagement are critical success factors for multilingual conversational AI, with voice-based interaction, native language support, and empathetic communication associated with higher patient satisfaction. However, trust scores remain variable (60–85%), particularly for diagnostic recommendations, suggesting that patients maintain appropriate skepticism about AI capabilities.

Recent literature emphasizes the importance of robustness, uncertainty management, and validation in the responsible deployment of medical AI technologies [29]. Our review shows that current multilingual conversational AI systems encounter significant challenges in these areas, including limited real-world validation; inadequate focus on uncertainty quantification; and insufficient assessment of robustness across linguistic variations, dialects, and code-mixing.

Compared to reviews of monolingual healthcare chatbots, this review reveals unique challenges related to language diversity, including English-language bias in LLMs [45], code-mixing failure modes [23], dialect variation, and cultural appropriateness. These challenges underscore the need for multilingual-specific evaluation frameworks and validation approaches.

4.9. Implications for Practice

Several implications emerge for clinical practice.

1. *Appropriate positioning:* Current multilingual conversational AI systems are most appropriately positioned as decision-support tools, triage assistants, or documentation aids rather than as autonomous diagnostic systems. Physician oversight remains essential.
2. *Language-appropriate care:* Healthcare organizations serving linguistically diverse populations should consider multilingual conversational AI as one component of a comprehensive language access strategy, complementing professional medical interpreters and bilingual providers.
3. *Patient-centered implementation:* Successful implementation requires a patient-centered design, including voice-based interaction, culturally appropriate communication, clear

explanations of system capabilities and limitations, and mechanisms for escalation to human providers.

4. *Workflow integration*: Integration into existing clinical workflows should prioritize the reduction of documentation burden and enhancement of patient–provider communication rather than the replacement of clinical judgment.
5. *Continuous monitoring*: Ongoing monitoring of system performance, patient outcomes, and equity impacts is essential, particularly for underrepresented languages and populations.

4.10. Implications for Policy

For policymakers and healthcare system leaders, key implications include:

1. *Regulatory Frameworks*: Develop clear, evidence-based regulatory frameworks for AI-based medical devices that balance innovation with patient safety [53].
2. *Reimbursement Models*: Consider reimbursement models that incentivize the adoption of technologies that improve efficiency and health equity.
3. *Standards Development*: Support development of standardized evaluation frameworks, outcome measures, and interoperability standards for multilingual healthcare AI.
4. *Bias and Fairness*: Regulatory frameworks should require assessment and mitigation of bias, with particular attention to underrepresented languages and populations [54].
5. *Data Governance*: Establish clear data governance frameworks addressing privacy, security, and ethical use of patient data in AI systems.

4.11. Future Research Directions

Several critical research gaps and future directions emerge from this review:

1. *Real-world comparative effectiveness studies*: Rigorous, real-world comparative effectiveness studies are urgently needed to evaluate diagnostic accuracy, clinical outcomes, workflow efficiency, and cost-effectiveness compared to those of standard care [55].
2. *Diverse population validation*: Validation studies in diverse, real-world populations are essential to ensure equitable performance across linguistic, cultural, and demographic groups.
3. *Code-mixing and dialect evaluation*: Systematic evaluation of system performance with code-mixed input, regional dialects, and non-standard language use is needed to assess real-world applicability.
4. *Long-term outcomes*: Long-term studies evaluating impacts on health outcomes, health disparities, patient–provider relationships, and clinician burnout are needed.
5. *Implementation science*: Research on implementation strategies, workflow integration, clinician acceptance, and organizational factors affecting adoption is critical for translating technical capabilities into clinical impact.
6. *Bias and fairness*: Systematic research on bias detection, measurement, and mitigation in multilingual healthcare AI is needed, with particular attention to underrepresented languages and populations.
7. *Standardized evaluation frameworks*: The development of standardized evaluation frameworks and outcome measures for multilingual conversational AI in healthcare would enable more rigorous comparison across systems and studies.
8. *Human–AI collaboration*: Research on optimal models of human–AI collaboration, including appropriate levels of automation, physician oversight, and patient autonomy, is needed.

4.12. Limitations

This systematic review has several important limitations:

1. *Heterogeneity and meta-analysis*: Substantial heterogeneity in study designs, populations, interventions, and outcome measures precluded meta-analysis. The findings are synthesized narratively, limiting the quantitative conclusions.
2. *Evidence certainty*: The overall evidence certainty is low to moderate, reflecting the predominance of prototype and proof-of-concept studies (65%), with limited real-world validation. Only 8% of studies reported comparative evaluations.
3. *Non-peer-reviewed sources*: Despite initial exclusion criteria, 8 of 49 included studies (16%) were non-peer-reviewed sources (arXiv preprints, Figshare/Zenodo deposits, OSF preprints). Two of the included studies were published in non-indexed regional journals (IJRASET and IRJAEH), which are not currently indexed in major academic databases; their inclusion reflects the limited availability of peer-reviewed evidence for the language contexts studied. All non-peer-reviewed sources were included when they represented the sole available evidence for specific language groups or underserved populations. This inclusion represents a methodological limitation and may affect the reliability of the findings.
4. *Google Scholar reproducibility*: Google Scholar was used as a supplementary database to capture gray literature and preprints. Due to its non-reproducible nature (personalized, location-dependent results), the search cannot be independently replicated. The primary systematic databases were Scopus, Web of Science, and PubMed.
5. *Absence of formal GRADE assessment*: Formal GRADE assessment was not applied due to the heterogeneity of study designs and the predominance of non-comparative studies. Evidence certainty was assessed narratively, and this method may be less rigorous than formal GRADE evaluation.
6. *Inter-rater reliability*: Kappa statistics were not formally calculated for inter-rater agreement at screening, data extraction, or risk of bias assessment stages. While all disagreements were resolved through consensus discussion, the absence of formal reliability metrics represents a methodological limitation.
7. *Data extraction verification*: Only 20% of data extractions were independently verified by a second reviewer (D.V.). While discrepancies were resolved through discussion, the absence of complete dual-reviewer extraction represents a methodological limitation that may affect the reproducibility and reliability of the extracted data.
8. *Language bias*: The review was limited to studies published in English, potentially excluding relevant research published in other languages.
9. *Publication bias*: The strong positive publication bias (96% of studies reporting favorable results) suggests potential selective publication of positive findings.
10. *Rapid field evolution*: The field of multilingual conversational AI is evolving rapidly. Findings may become outdated quickly as new technologies and systems are developed.
11. *Limited long-term data*: Most studies reported short-term outcomes. Long-term impacts on health outcomes, health disparities, and clinical practice remain unknown.

5. Conclusions

This systematic review synthesized evidence from 49 studies on multilingual conversational AI chatbots for case history-taking and patient communication in healthcare. The review identified substantial growth in this field, with systems supporting 42 distinct languages across diverse healthcare domains. Individual system evaluations reported promising diagnostic accuracy (72–92% in heterogeneous evaluations), workflow efficiency gains (30–70% time savings in specific contexts), and positive patient engagement (75–92% satisfaction). However, the overall evidence certainty was low to moderate, reflecting the predominance of prototype and proof-of-concept studies with limited real-world validation.

Multilingual conversational AI systems demonstrate potential to advance health equity by reducing language barriers and expanding access to care for underserved populations. However, critical challenges remain, including English-language bias in medical LLMs, code-mixing failure modes, limited real-world validation, workflow integration barriers, and concerns about bias and fairness.

Current systems are best viewed as decision-support tools rather than as fully autonomous diagnostic systems. Multilingual AI should serve as a supplement to, not a replacement for, professional medical interpreters and bilingual providers. Clear regulatory pathways and reimbursement mechanisms are necessary. Continuous monitoring of performance, outcomes, and equity impacts is crucial.

With continued research, rigorous validation, and attention to equity and implementation challenges, multilingual conversational AI has the potential to meaningfully improve healthcare delivery and reduce language-related health disparities.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/informatics13080129/s1>, Table S1: Complete search strategies for all databases; Table S2: Data extraction form; Table S3: Risk of bias assessment details; Table S4: PRISMA 2020 checklist; Table S5: Comprehensive summary of all 49 included studies. Refs. [1–24,30–44,46,47,49–51,56–61] can be found in the Supplementary Materials.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
BERT	Bidirectional Encoder Representation Form Transformers
GPT	Generative Pretrained Transformer
LLMs	Large Language Models
NLP	Natural Language Processing
RAG	Retrieval Augmented Generation
L2M3	Multilingual Medical Large Language Model
EHR systems	Electronic Health Record
ROBINS-I	Risk of Bias in Non-Randomized Studies of Interventions
AMSTAR-2	A Measurement Tool to Assess Systematic Reviews

PROs	Patient-reported outcomes
MT	Machine Translation
NMT	Neural Machine Translation
GRADE	Grading of Recommendations, Assessment, Development and Evaluations
AUC-ROC	Area Under the Receiver Operating Characteristic Curve

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